# The difficulty of diagnostics

**ROC curves and thresholds in policy & public health**

## Making black-and-white decisions in a grayscale world

The world we live in may have absolute, black and white truths, indeed we have built our social and judicial systems on this premise—an individual is guilty or innocent of a crime, needs public assistance or is trying to cheat the system, is merely exercising protected 1st or 2nd amendment rights or is threatening others' safety—but finding that clear line between the two is pretty damn tricky. We are almost always forced to use imperfect information to delineate these binary outcomes and no matter how black and white the reality may be, there are broad swaths of gray in the decision making.

Many years ago I spent an enlightening year as an AmeriCorp VISTA member helping administer a private energy assistance program. My job was to help people in need keep the power or heat on in their houses, mostly by helping pay down their overdue balances with money from private donations. I looked at individual applications, talked to them, and then decided whether we could help them and how. As a 22 year old college graduate, I had very little experience with people on the edge. I was often overwhelmed having to determine whether they were telling the truth, or whether they could follow through on their promises and payment plans I had arranged. It was gray as far as the eye could see. I was always a little relieved when people were clearly above the income cutoff; while I often felt bad that we couldn't help them, I wasn't forced to make a judgment about a person or their plight.

The problem was, I felt damned no matter what I did. If I was too generous with assistance, I was surely helping people that were milking, if not cheating the system. Moreover, there was a real chance of not having enough money for the truly needy by the end of the season. On the other hand, if I were too skeptical or stringent, only helping where I felt certain that they were under water through no (or little) fault of their own, and that they weren't coming to depend on our assistance, well then I was almost certainly excluding people that needed our help, but had a harder time making their case. I frequently sought the advice of the more seasoned women who ran other public programs, but learned I was going to quickly become cynical or apathetic if I kept trying to make these important decisions with very imperfect data. So I applied to graduate programs in biology in search of Truth[1], or at least unambiguous facts and rigorously tested hypotheses.

[1] With a capital "T".

The irony is that that gray area of decision making is rampant in biology. Well, it's common in science in general, but maybe especially in my chosen field of disease ecology. The difference is, I think, that in the sciences we are encouraged to be open about the uncertainty, about the gray areas, and also that we have some useful tools to deal with it.

## Is that an infection, or glowing lint?

Consider the most essential aspect of a study about infectious disease, determining whether an individual is infected or not. The reality is that an individual falls into one category or the other, there are no "sort of" infected individuals[2], but we cannot know an individual's state with complete certainty[3]. Instead, we measure something directly or indirectly related to the infection such as the worms in someone's poo[4], the presence or quantity of pathogen DNA, or the degree to which antibodies in our blood react to a pathogen. Let us stick with antibodies.

[2] Though there are large differences in the *intensity* of infections, disease symptoms, etc.

[3] At least not without experimentally infecting them!

[4] Yup! There's a person who's job is to look in those stool samples and count worms. It's actually pretty neat!

If you've been infected with a pathogen and have recovered you have probably developed antibodies to many parts of that pathogen. So when someone draws your blood they can see if you have antibodies in it that react to those pathogen parts, called epitopes. If so, there's some signal, usually a color change depending on how the test is put together. But you might have a little signal while I produce a big one.

For instance, individuals that were recently infected, are immunosuppressed, or are just, in the words of a friend and colleague, "wonky," and so they may be infected without producing many antibodies against the pathogen. Over all, there is a distribution of antibody reactions from infected individuals (Figure 1).

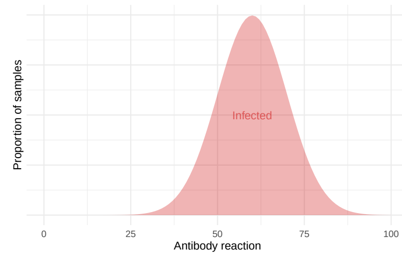


Figure 1: Hypothetical distributions of an antibody reaction (e.g., in an ELISA) from infected individuals.

You might expect all the samples from uninfected individuals would be very low, maybe even zero. However, you, me, a mouse, or a bird all produce antibodies that cross react with pathogens we've never seen. There is always a bit of a signal. So while the distribution of antibody reactions from uninfected individuals is *low*, it might still overlap a bit with the infected distribution (Figure 2). In other words, there is always a gray area (or purplish in Figure 2) of signals that could be either uninfected or infected individuals.

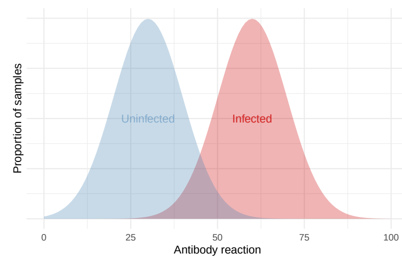


Figure 2: Distributions of an antibody reactions from infected *and* uninfected individuals.

Because we're trying to categorize individuals into infected and uninfected categories, we need to choose some threshold amount of signal (Figure 3) above which the test is scored as positive and below which it is negative. However, unless there is perfect separation[5] (i.e., the two curves do not overlap) there will be some **false positives** (uninfected samples that test positive) or some **false negatives** (infected samples that test negative). We can shift this threshold lower to minimize the number of false negatives, but this ends up increasing the number of false positives (compare Figure 4 with Figure 3)! Or we could increase our threshold to ensure there were very few false positives, but that would come at the cost of more false negatives. There is an inherent trade-off between increased **sensitivity**—ensuring there are few or no false negatives—and **specificity**—ensuring there are few or no false positives.

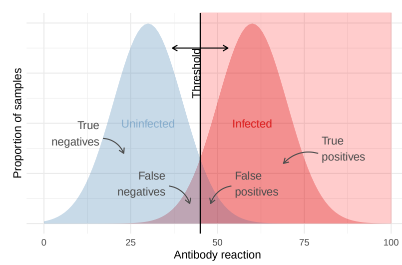


Figure 3: Above some threshold amount of antibody reaction a sample is scored as positive (pink background) and below it is negative.

The same is true of virtually every diagnostic method. One lab I worked in detected the agent of Lyme disease in black legged ticks using a reagent that made the bacterium fluoresce under a microscope. Unfortunately, pieces of lint also fluoresced and sometimes curled in just the right way to look like a bacterium.

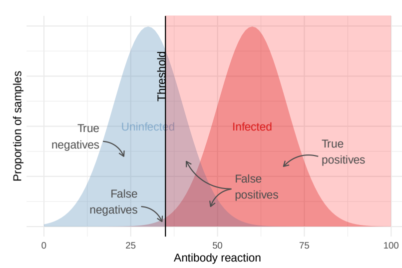


Figure 4: We can lower the threshold (i.e., be less stringent) so there are fewer false negatives, but this will produce more false positives.

[5] Better diagnostic test have more differentiation between the distributions, but we can rarely *completely* separate them.

And of course if you didn't look at the right place or blinked at the wrong time, you might even miss this glowing signal. So we undoubtedly scored some uninfected ticks as infected (false positives) and some infected ticks as uninfected (false negatives). There may be ways to narrow this gray area—use a better diagnostic test, use a two-step test with different methods[6]—but the gray area is almost universally present[7] whether in diagnostic tests or in benefit determinations.

[6] More on this later.

[7] Even physics has this problem. For instance, was that signal in the CERN accelerator really a new particle or just some noise from the detector or another particle or…? Their key advantage is that they can do the experiment a gazillion times, checking and re-checking for other sources of putative signals, until they are exceedingly confident that what they saw, and keep seeing, is real.

## Classification and ROC curves

So how do scientists deal with these gray areas? How do we decide where to draw the line between one category and another? Well, we acknowledge the trade-off between sensitivity and specificity and then find the threshold value that does the best job of achieving a balance between these competing interests given our goals.

One incredibly useful tool to think through these trade-offs is the ROC (Receiver operating characteristic) curve (Figure 5). This is the product of yet another detection problem, detecting enemy airplanes or ships using early versions of radar in World War II. The problem is the same as with detecting infection status, if not a bit more immediate. Is that blip I see on the screen a real plane or something else like a flock of birds or a phantom in the electronics? These curves help us see how good or bad our test might be and, perhaps even more importantly, better understand the trade-offs we face with imperfect tests.

Imagine we were to move the threshold in the previous figures from the far right (super stringent) to the far left (super lenient) and record the number or proportion of false negatives and false positives at each point. If we plotted 1 - false positive rate (aka sensitivity or the true positive rate) against the false positive rate (=1-specificity) we would have a nice curve sweeping an arc into the upper left corner. These curves describe the trade-off between sensitivity and specificity.

Notice the two thresholds we used above are plotted along this curve. The first was a threshold of 45, which resulted in a Sensitivity and Specificity of 0.933. That is, 93 out of 100

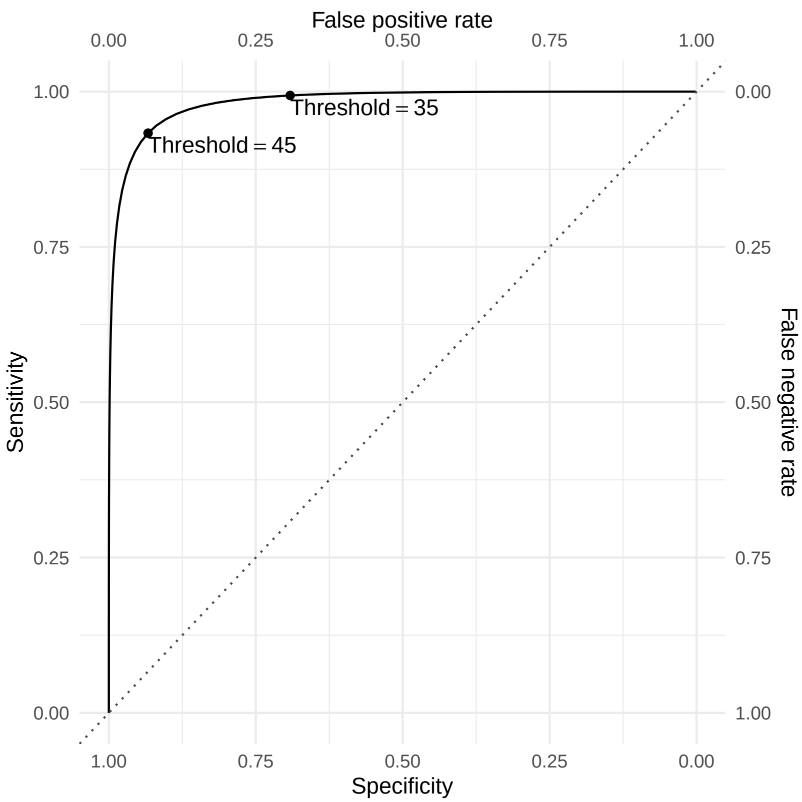


Figure 5: ROC plot for the hypothetical antibody test. The cutoffs from the prior figures are plotted along the curve. Notice the axis values are reverse from top to bottom and left to right. When a test is pefectly sensitive there are no false negatives.

truly infected samples will test positive and 93 out of 100 truly uninfected samples will test negative; but seven in each case will be scored wrong. If we were to reduce the threshold to 35 we would increase the sensitivity to 0.994—over 99 out of 100 truly infected samples will test positive—but this comes at the cost of many more false positive tests, 31 out of 100 truly _un_infected samples will test positive![8]

The beauty of these curves is that they help us visualize how the rates of false negatives and false positives change as you move the threshold from more stringent (no false positives, but many false negatives) to less stringent (more false positives, but fewer false negatives)[9]. The ideal diagnostic test or radar system would produce a curve that is essentially a rectangle[10] where there is a threshold value that perfectly delineated airplanes from birds, infected from uninfected individuals, the needy from the conniving. In this case there would be no overlap between the red and blue distributions in Figure 2.

However, in the real world there is almost always some noise. As the amount of noise increases (i.e., the red and blue distributions overlap more and more), the distinction between infected and uninfected, or plane and bird, becomes more and more ambiguous and the ROC curve approaches the one-to-one line (dotted line in Figure 5[11]). This is essentially the same as flipping a coin and saying if it's heads then the person is infected. Or more directly, the test is useless. Most of the time our diagnostic test or radar system falls somewhere in between, which means we have a trade-off between certainty that we will not miss an airplane or infection (low chance of false negatives) and an increase in false alarms (false positives).

## What is the right threshold? Is there a right threshold?

Where should a person draw the line, then? If you give the problem a bit of thought my guess is that you recommend finding the threshold that maximizes overall accuracy[12] or, equivalently, minimizes the total number of false positives and false negatives. More accurate[13] is better, right? Well yes…so long

[8] Heck! I could give you a diagnostic test with 100% sensitivity—a box that simply always said "infected" would work! Just don't ask about its specificity!

[9] Try this interactive app to get a better sense of how signal, noise, and thresholds interact.

[10] And the "area under the curve" (AUC), if you've run across this term, is =1.

[11] With an AUC of 0.5.

[12] Accuracy = $\frac{TP+TN}{\text{All tests}}$

[13] It is worth noting that if someone says a test is 97% accurate, you do not know if the inaccurate results are mostly false negatives or false positives or both. It is hard to summarize a test's performance in one single number!

as the costs of false negatives and false positives are the same. If you were in charge of the radar system in WWII would you be more worried about the cost of scrambling your forces for no good reason in response to a false positive or about missing the first airplanes in an invasion due to overly stringent criteria and false negatives? Would you want the threshold that maximized accuracy or would you rather move that threshold a bit to ensure you had very, very few false negatives at the cost of perhaps a lot more false positives? The cost of scrambling your airplanes to intercept a phantom invader is not negligible, but is probably quite small relative to the cost of an enemy airplane going undetected in which case you might lose lives, a defensive post, or a battle. And so you will probably choose the less stringent threshold and a lot of false positives. In peacetime, however, your calculations probably change.

The same is true when it comes to diagnostic tests. There is a debate raging over advice on who should get a mammogram and when. Epidemiologists have calculated that the combined costs of all of the false positive diagnoses—the unneeded treatments and surgery, not to mention the worry and expenses—outweigh the benefits of subjecting nearly all women over a certain age to the test in order to minimize the chances that a malignant tumor goes undetected. Or similarly, was it better to tweak the test to ensure we detected as many COVID-19 cases as possible, but inadvertently caused a bunch of uninfected people to have to quarantine themselves, or was it better to be able to trust that everyone who tested positive was really positive[14]?

[14] One way to express the utility of a test is the positive predictive value (=TP/(TP+FP)), sometimes called precision, which basically tell us the likelihood that a positive test is actually a true positive, and, similarly, the negative predictive value (=TN/(TN+FN)), which is the reverse.

By focusing on just one type of cost (or one axis of the ROC curve) I think we miss the larger point that there are trade-offs and that the relative costs and benefits are, at least in principle, calculable. We may disagree on how to value different outcomes, but the underlying trade-off between false positives and false negatives is always there in the real world but is, in principle, objective. I think that being transparent about why we choose particular cutoffs, and what that means in terms of trade-offs, would be useful to our society more generally. At a minimum, though, I think that ROC curves, and the perspectives embedded in them, can help us have more productive discussions about where we should draw the lines.